

ICRAR Research Project Report

Computer Simulations of Globular Cluster Formation in The Early LMC

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Abstract

NGC 1876 is one of the galactic globular clusters (GCs) that have been observed to show an Mg-K anticorrelation. The physical mechanism responsible for this pattern remains poorly understood and is still the subject of active debate. In order to understand the physical mechanisms responsible for the Mg-K anti-correlation, we have performed hydrodynamical simulations of GC formation in the early LMC. In this work, we qualitatively reproduce the observed Mg-K anticorrelation using chemical yields of super AGB stars predicted by Ventura et al. (2012). Within the simulation, we identify a globular cluster analogue with a compact half-mass radius of $R_{1/2} = 2.44$ pc and a radial density profile qualitatively similar in shape to NGC 1786, and an enclosed mass within 50 pc $M(R < 50 \text{ pc}) = 2.83 \times 10^6$ that is higher than literature estimates for NGC 1786 by about an order of magnitude.

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1 Introduction

A Mg–K anticorrelation was first observed in NGC 2419 by Mucciarelli et al. (2012) and Cohen et al. (2012). In this cluster, about one third of the cluster mass belongs to a Mg-poor, K-rich population relative to the Mg-normal population (Cohen et al. 2012). Ventura et al. (2012) proposed that this could arise if second-generation stars formed from material processed by hot bottom burning (HBB) in massive asymptotic giant branch (AGB) stars and super-AGB (SAGB) stars. In this framework, the most Mg-poor stars are expected to be strongly oxygen-depleted and not strongly sodium-enhanced (Ventura et al. 2012). Iliadis et al. (2016) used reaction-network calculations to constrain the temperature and density conditions needed to reproduce the observed Mg–K (and related) abundance patterns, and noted that improved oxygen measurements could provide additional constraints on the proposed polluter sites. However, oxygen measurements in NGC 2419 remain too sparse to draw firm conclusions (Cohen et al. 2012).

In NGC 2808, Mucciarelli et al. (2015) identified four Mg-deficient stars, all of which are K-enhanced, supporting the existence of a Mg–K anticorrelation in another Globular Cluster (GC). They also reported a maximum oxygen depletion of $\simeq -1.3$ dex, which is comparable to the $\simeq -1.7$ dex depletion predicted by the models of Ventura et al. (2012). Alvarez Garay et al. (2022) found that ω Centauri also shows a clear Mg–K anticorrelation and reported a super K-rich star (43241_J10) with $[K/Fe] = +1.6$ dex; they suggested that this star may have formed from AGB and SAGB ejecta with little or no dilution by pristine gas.

The first extragalactic globular cluster reported to host a Mg–K anticorrelation is NGC 17860.1– M_{upp} and write it (as implemented in our sweep) as a single power law

in the Large Magellanic Cloud Alvarez Garay et al. (2025). They argue that the Mg–K anticorrelation is a characteristic of old, massive, metal-poor globular clusters that contain a Mg-poor subpopulation, and is largely independent of the host galaxy environment (Alvarez Garay et al. 2025). NGC 1786 also exhibits the canonical light-element anticorrelations associated

with CNO-cycle processing and the NeNa and MgAl proton-capture chains (Alvarez Garay et al. 2025).

A leading explanation for the Mg–K anticorrelation is the AGB/sAGB polluter scenario. In this picture, hot bottom burning in massive AGB and super-AGB stars can activate the MgAl cycle, depleting Mg, and can also produce K via proton captures on Ar at sufficiently high temperatures. The processed material is then expelled in stellar winds and mixes with pristine gas in the intracluster medium. Second-generation stars forming from this mixture inherit low Mg and enhanced K, producing an Mg–K anticorrelation (D’Ercole et al. 2010).

We outline this work as follows. In Section 2 we describe the model assumptions and simulation setup. In Section 3 we describe the analysis methodology used to measure chemical and structural properties of candidate clusters. In Section 4 we present the results for our candidate GC analogue. In Section 5 we discuss interpretation, limitations, and key uncertainties, and we conclude in Section 6.

2 Model and Simulation

2.1 Estimation of initial GMC mass

We adopt a present-day mass of $3.72 \times 10^5 M_{\odot}$ for NGC 1786 (Vanaraj et al. 2021). Following the mass-budget convention in Decressin et al. (2010), we treat the present-day observable population as long-lived low-mass stars with $m = 0.1\text{--}0.8 M_{\odot}$.

To relate the present-day globular cluster mass in long-lived stars to its initial stellar mass, we approximate the IMF as a single-slope power law (Kroupa et al. 2024) over $m =$

$$\xi(m) \propto m^{-\alpha}, \quad (1)$$

where α is varied in our parameter exploration. We define f_{low} as the fraction of the initial stellar mass that forms low-mass, long-lived

stars ($0.1\text{--}0.8 M_\odot$):

$$f_{\text{low}} = \frac{\int_{0.1}^{0.8} m \xi(m) dm}{\int_{0.1}^{M_{\text{upp}}} m \xi(m) dm}. \quad (2)$$

We then parameterise dynamical mass loss (e.g. tidal stripping) with a stripping fraction f_{st} , such that

$$M_{\text{GC},p} = M_{\text{GC},i} f_{\text{low}} (1 - f_{\text{st}}), \quad (3)$$

where $M_{\text{GC},i}$ is the initial stellar mass of the cluster and $M_{\text{GC},p}$ is the present-day mass in surviving low-mass stars.

The current observed power-law slope for NGC 1786 is -1.16 (Baumgardt n.d.). To estimate the stripping fraction in the low-mass population using the same mass range as above, we model the present-day mass function as

$$\xi_{\text{obs}}(m) \propto m^{-\alpha_{\text{obs}}}, \quad \alpha_{\text{obs}} = 1.16, \quad (4)$$

and (as implemented) assume the same overall normalisation for ξ_{obs} and ξ so the comparison depends only on the slopes. We then estimate the retained fraction of low-mass *mass* over $0.1\text{--}0.8 M_\odot$ via

$$1 - f_{\text{st}} = \frac{\int_{0.1}^{0.8} m \xi_{\text{obs}}(m) dm}{\int_{0.1}^{0.8} m \xi(m) dm}, \quad (5)$$

so that f_{st} represents the fraction of low-mass material lost due to stripping.

The original GMC mass $M_{\text{GMC},i}$ is given by

$$M_{\text{GC},i} = \epsilon M_{\text{GMC},i}, \quad (6)$$

where ϵ is the star formation efficiency.

We swept over the upper-mass cutoff M_{upp} ($8\text{--}50 M_\odot$) and the IMF slope α ($1.1\text{--}3.0$) (Fig. 1). We find that M_{upp} does not change the inferred GMC cutoff mass significantly beyond a moderate value, while the dominant sensitivities come from the star formation efficiency ϵ and the IMF slope α . From this sweep, a cutoff mass of order $10^7 M_\odot$ is reasonable.

2.2 Hydrochemical simulation

We use a revised version of the hydrochemical simulation code of Bekki (2013). We performed

Table 1: Code-unit scalings adopted for the two simulations.

Quantity	Host-galaxy run	Zoom-in run
Mass unit, M_0	$6 \times 10^{10} M_\odot$	$10^6 M_\odot$
Length unit, L_0	17.5 kpc	100 pc
Time unit, T_0	0.14 Gyr	1.49×10^7 yr

Table 2: Host galaxy total masses used in the initial conditions.

Component	Total mass ($M_\odot$)
Gas	1.62×10^{10}
Stars	1.80×10^9
Dark matter	1.002×10^{12}

two simulations: (i) a host-galaxy-scale run to form and identify the target GMC, and (ii) a parsec-scale zoom-in run of the selected GMC at higher mass resolution.

The host-galaxy initial conditions are specified in physical units ($M_\odot$). Table 2 summarises the total stellar, gas, and dark matter masses adopted for the host galaxy model.

The two runs use different code-unit scalings, summarised in Table 1.

In the host-galaxy run, we select the GMC using a distance-based particle selection criterion (membership determined from inter-particle separations). We then re-simulate the selected GMC in a zoom-in calculation using particle splitting: each parent gas particle is replaced by 300 child gas particles, and each parent star particle is replaced by 10 child star particles, while conserving the parent particle mass and centre-of-mass (COM) properties.

The particle masses in the zoom-in run are summarised in Table 3.

In the zoom-in run, time integration uses $dt = 5 \times 10^{-4}$ with a minimum allowed step $dt_{\text{min}} = 10^{-4}$ in code units, corresponding to $dt \approx 7.45 \times 10^3$ yr and $dt_{\text{min}} \approx 1.49 \times 10^3$ yr.

3 Analysis Methodology

3.1 GC Physical Characteristics

To characterise the Mg–K abundance distribution in each model cluster, we selected star particles within 50 pc of the adopted cluster centre and plotted $[\text{Mg}/\text{Fe}]$ against $[\text{K}/\text{Fe}]$. With 1 simulation length unit = 100 pc, this corre-

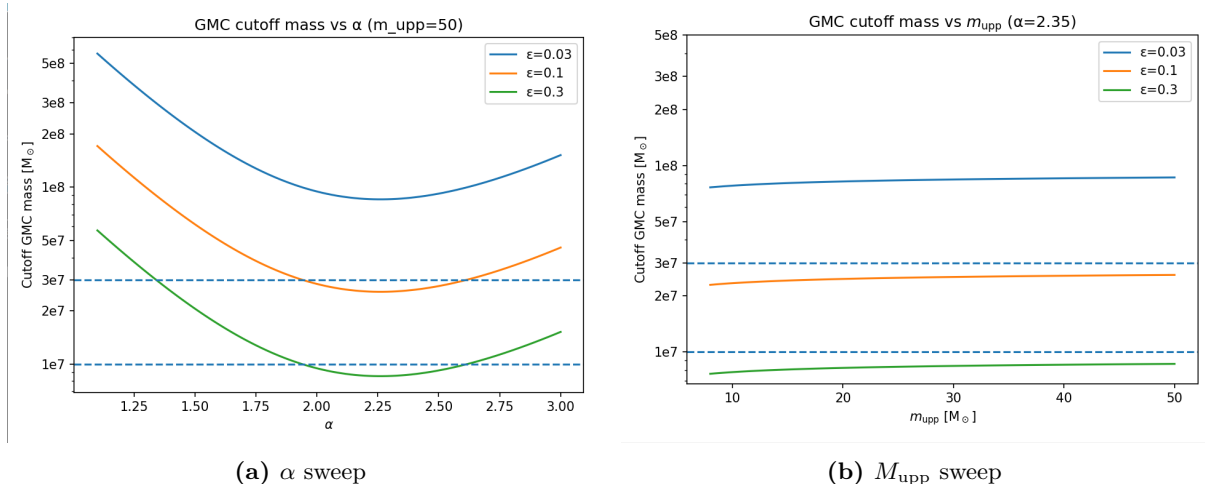


Figure 1: Parameter sweeps showing the sensitivity of the inferred GMC mass scale to α and M_{upp} .

Table 3: Particle mass resolution adopted in the zoom-in run.

Particle population	Symbol	Mass (code units)	Mass (physical units)
Gas	m_{gas}	$1.80000 \times 10^{-5} M_0^{\text{zoom}}$	$18 M_{\odot}$
Newly formed stars	m_{new}	$1.80000 \times 10^{-5} M_0^{\text{zoom}}$	$18 M_{\odot}$
Pre-existing (old) stars	m_{old}	$5.40000 \times 10^{-4} M_0^{\text{zoom}}$	$540 M_{\odot}$

sponds to $R_{\text{cut}} = 0.5$ in simulation units. For each star particle, we used the simulation mass fractions X_{Mg} , X_{Fe} , and X_{K} and computed

$$[\text{Mg}/\text{Fe}] = \log_{10} \left(\frac{X_{\text{Mg}}}{X_{\text{Fe}}} \right) - \log_{10} \left(\frac{X_{\text{Mg}}}{X_{\text{Fe}}} \right)_{\odot}$$

$$[\text{K}/\text{Fe}] = \log_{10} \left(\frac{X_{\text{K}}}{X_{\text{Fe}}} \right) - \log_{10} \left(\frac{X_{\text{K}}}{X_{\text{Fe}}} \right)_{\odot},$$

adopting solar mass fractions $X_{\text{Fe},\odot} = 1.17 \times 10^{-3}$, $X_{\text{Mg},\odot} = 5.15 \times 10^{-4}$, and $X_{\text{K},\odot} = 3.649 \times 10^{-6}$. We distinguish new stars (formed during the simulation) from old stars (present in the initial conditions) and compare the simulated distribution with observed abundances including measurement uncertainties.

We computed radial mass profiles of the stellar component after recentering each snapshot on a robust density centre. The centre was determined in two stages: (i) a k-nearest-neighbour density-peak estimate using a KD-tree (the distance d_k to the k -th nearest neighbour, with $k = 64$), followed by a local mass-weighted refinement; and (ii) a shrinking-sphere iteration within a fixed aperture to obtain a converged mass-weighted centre. After recentering, radii were computed in 3D and converted using 1 simulation length

unit = 100 pc, and stellar masses were converted via $m_{\star}(M_{\odot}) = m_{\star}(\text{sim}) \times 10^6$. Within $R_{\text{max}} = 50$ pc, we constructed the enclosed mass profile $M(< R)$ by sorting stars by radius and taking the cumulative sum, and estimated the half-mass radius $R_{1/2}$ by solving $M(< R_{1/2}) = \frac{1}{2} M(< R_{\text{max}})$ with linear interpolation. We also computed the three-dimensional radial mass density profile $\rho(r)$ by binning stellar mass into $N_r = 500$ spherical shells and dividing by the shell volumes $V_i = \frac{4\pi}{3} (r_{i+1}^3 - r_i^3)$, reporting profiles separately for new and old stars as well as the total.

In addition, we computed projected stellar surface density maps by projecting recentered positions onto the (x, y) plane, binning stellar mass into a uniform $N_{xy} \times N_{xy}$ grid over a square field of view, and converting mass-per-pixel to surface density via $\Sigma_{ij} = M_{ij}/(\Delta x \Delta y)$ in units of $M_{\odot} \text{ pc}^{-2}$. In our implementation we used $N_{xy} = 300$ (i.e. 300×300 pixels), with $\Delta x = (x_{\text{max}} - x_{\text{min}})/N_{xy}$ and $\Delta y = (y_{\text{max}} - y_{\text{min}})/N_{xy}$ set by the chosen field of view.

3.2 Consistency Checks

We performed consistency checks by applying our analysis pipeline to two globular clusters (GC 1 and GC 2) drawn from the same simulation suite. In both models, the newly formed stars populate a non-smooth, clumpy locus in the $[\text{Mg}/\text{Fe}]$ – $[\text{K}/\text{Fe}]$ plane (Fig. 2). A clumpy

distribution is qualitatively consistent with scenarios in which enriched stars form from gas that is not perfectly mixed cluster-wide, but instead reflects spatially and temporally variable dilution between super-AGB (sAGB) ejecta and more pristine material. Such dilution-driven enrichment, and the expectation that processed material can be retained and recycled into later star formation, are common elements of AGB/sAGB multiple-population models (D’Ercole et al. 2008; Calura et al. 2019; Gratton et al. 2012). Discrete sub-populations (appearing as clumps in chemical or photometric planes) are also observed in several Galactic globular clusters, motivating a comparison between clumpy model predictions and clumpy data products (Milone et al. 2017; Kenji Bekki et al. 2017).

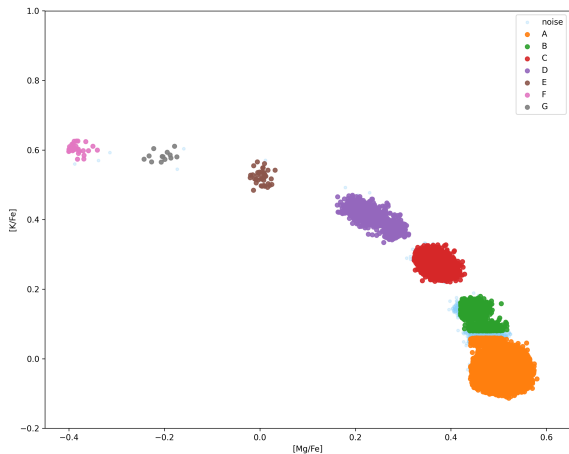


Figure 2: $[\text{Mg}/\text{Fe}]$ – $[\text{K}/\text{Fe}]$ distribution for two model globular cluster, showing a clumpy anticorrelation pattern.

To interpret these clumps, we assign stars into groups (A–G) ordered by increasing degree of enrichment: group A is consistent with the initial (GMC) composition (centred on $[\text{K}/\text{Fe}] \approx 0$ and $[\text{Mg}/\text{Fe}] \approx 0.5$), while groups B–G show progressively lower magnesium and higher potassium. This grouping is conceptually analogous to the widely used separation of multiple populations into “primordial” and increasingly “processed” components in other abundance planes (for example in Na–O), which is commonly interpreted as reflecting different levels of dilution between processed ejecta and pristine gas (Carretta et al. 2009).

We then compare the chemical grouping to

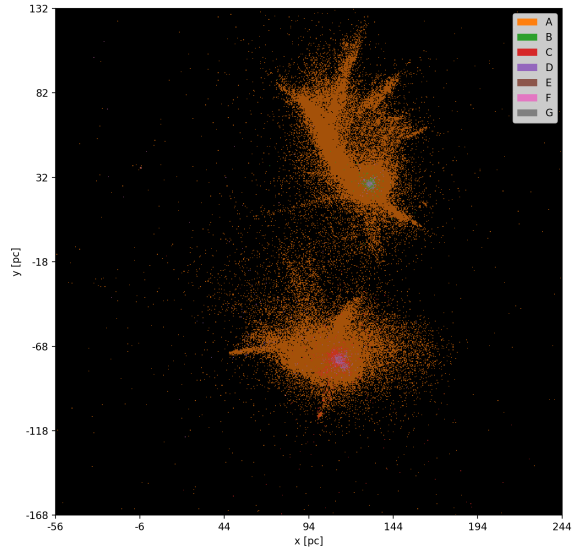


Figure 3: Spatial distribution of stars coloured by chemical group (A–G) for the two model globular clusters.

the spatial distribution of stars (Fig. 3). We find that the more strongly enriched groups are preferentially concentrated toward the cluster centre, and in both models the central regions are dominated by group D. Central concentration of enriched populations is a generic prediction of self-enrichment models in which slow AGB/sAGB winds are retained and accumulate in the inner regions before forming later-generation stars (D’Ercole et al. 2008; Calura et al. 2019). Over time, two-body relaxation mixes populations and can reduce (but not necessarily erase) initial radial differences (Vesperini et al. 2013); observationally, second-generation stars are often found to be more centrally concentrated than first-generation stars, particularly in systems that are not fully spatially mixed (Lardo et al. 2011).

Finally, the intermediate islands differ between the two clusters: group B is present only in one model, while group C is present only in the other, whereas the baseline component (group A) is present in both. This indicates that the detailed set of intermediate compositions can vary between clusters, consistent with cluster-to-cluster differences in enrichment and dilution histories.

We characterise the stellar structure using both a projected surface-mass-density map and a spherically averaged three-dimensional mass-

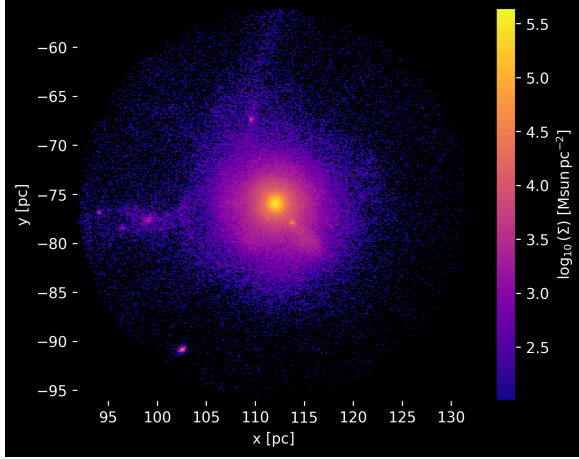


Figure 4: Projected surface-mass-density map of GC 2 for newly formed stars, $\Sigma(x, y)$, computed on a 300×300 grid.

density profile, each recentered on the robust density centre described above. The projected map is computed for the newly formed stellar population only, by binning stellar mass onto a 300×300 Cartesian grid in the snapshot plane to obtain $\Sigma(x, y)$ in units of $M_{\odot} \text{pc}^{-2}$. In the displayed field, the map spans approximately $\Sigma \sim 10^2$ – $10^{5.5} M_{\odot} \text{pc}^{-2}$. The radial profile is computed in three dimensions for the total, new, and pre-existing stellar components by binning stars into 500 spherical shells about the density centre, yielding $\rho(r)$ in units of $M_{\odot} \text{pc}^{-3}$; over the plotted radii we find $\rho \sim 10^0$ – $10^5 M_{\odot} \text{pc}^{-3}$.

The normalisations of Σ and ρ are not directly comparable because they have different dimensions: the surface density is the line-of-sight integral of the volume density,

$$\Sigma(R) = \int_{-\infty}^{+\infty} \rho(R, z) dz. \quad (7)$$

For this snapshot, the fact that Σ_{max} and ρ_{max} differ by less than an order of magnitude is consistent with an effective line-of-sight depth of order a few parsecs through the central region. Peak central densities of order $\rho \gtrsim 10^5 M_{\odot} \text{pc}^{-3}$ are also comparable to those reached by compact newly formed subsystems in hydrodynamical proto-GC simulations (Calura et al. 2019).

Both the projected map and the radial profile are sensitive to binning choices. In general, histogram and grid-based density estimates

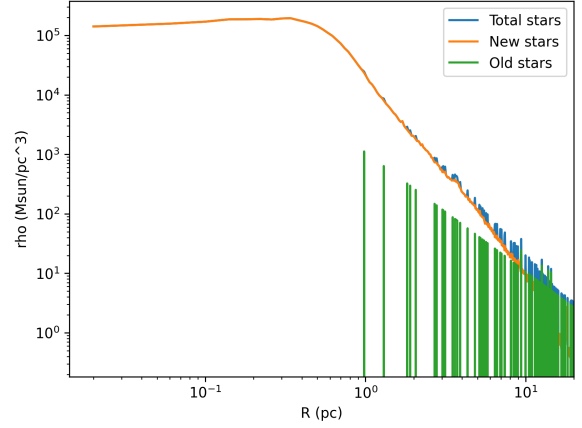


Figure 5: Spherically averaged three-dimensional stellar mass-density profile of GC 2, $\rho(r)$, shown separately for new stars, old stars, and the total.

depend on bin scale and bin placement, and astronomy commonly uses adaptive binning or alternative tessellations to reduce artefacts when data are sparse or strongly centrally concentrated (Knuth 2019; Cappellari et al. 2003; Diehl et al. 2006). This provides a plausible methodological contribution to differences between (i) a spherical-shell estimate of $\rho(R)$ and (ii) a pixelised estimate of $\Sigma(x, y)$, in addition to genuine projection effects.

The surface-density map shows one dominant central concentration plus multiple smaller overdensities distributed across the field. Because this map traces only newly formed stars, these substructures naturally highlight where recent star formation occurred. More broadly, spatial substructure is expected in dynamically young, assembling stellar systems and is erased on short timescales as the system relaxes and mixes (Goodwin et al. 2004; Allison et al. 2010; Parker et al. 2014; Portegies Zwart et al. 2010). In the radial profiles, the total and new-star components show an overall decline with radius, while the old-star profile becomes increasingly noisy at large R , consistent with low particle counts per shell in the outskirts. At small radii ($R \lesssim 1 \text{ pc}$ in the plotted profile) the density flattens, which we interpret as resolution-limited: inside a scale set by the force softening and/or the minimum hydrodynamic smoothing length, inner density slopes are not expected to be numerically converged (Power et al. 2003).

Qualitatively, the projected concentration and declining radial trend are consistent with the centrally concentrated profiles commonly used to describe globular clusters in projection (King 1966; McLaughlin et al. 2005).

4 Results

Here we present the candidate globular cluster (GC) model and summarise its structural and chemical properties, including the projected surface-mass-density map, the spherically averaged radial mass-density profile, the $[\text{Mg}/\text{Fe}]$ – $[\text{K}/\text{Fe}]$ distribution, and basic integrated quantities (enclosed mass, half-mass radius, and the number of star particles).

Figure 6 shows that the projected distribution of newly formed stars is noticeably non-circular, with an elongated central structure. The map also shows a secondary overdensity at lower surface density, consistent with substructure in the outer regions of the selected aperture.

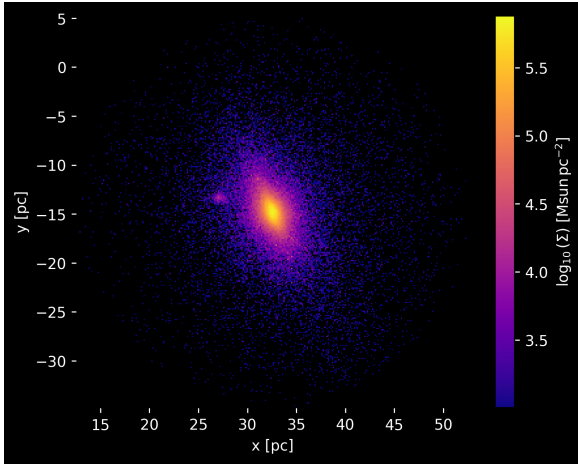


Figure 6: Projected surface-mass-density map of the candidate model for newly formed stars, $\Sigma(x, y)$, computed on a 300×300 grid.

Figure 7 shows the spherically averaged three-dimensional stellar mass-density profile spanning approximately 10^5 to 10^{-1} in $\rho(r)$. The inner profile flattens for $r \lesssim 1$ pc, which is consistent with finite mass and spatial resolution in the central region. At $r \gtrsim 10$ pc, the new-star component shows an excess relative to a smooth decline, consistent with the outer substructure visible in Figure 6. At large radii,

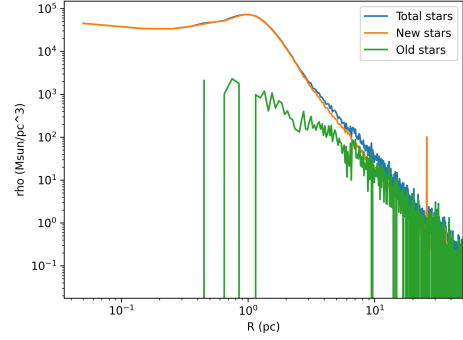


Figure 7: Spherically averaged three-dimensional stellar mass-density profile of the candidate GC, $\rho(r)$, shown separately for new stars, old stars, and the total.

the old-star profile becomes sparsely sampled, producing bins with low or zero counts.

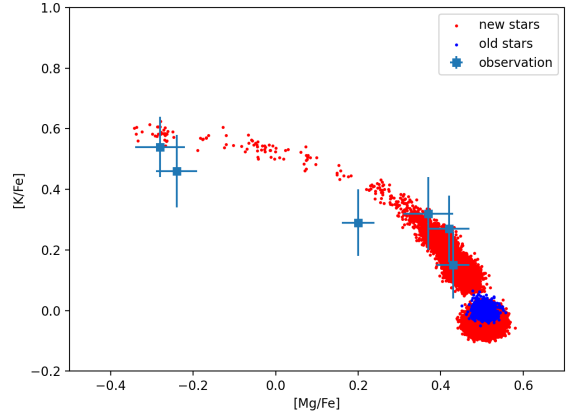


Figure 8: $[\text{Mg}/\text{Fe}]$ – $[\text{K}/\text{Fe}]$ distribution for the candidate model GC, showing an anticorrelation pattern.

Figure 8 shows a clear $[\text{Mg}/\text{Fe}]$ – $[\text{K}/\text{Fe}]$ anticorrelation in the selected stars. Qualitatively, the distribution lies within the observational uncertainty envelope reported for NGC 1786 by Alvarez Garay et al. (2025).

The enclosed stellar mass within 50 pc is $M(< 50 \text{ pc}) = 2.83 \times 10^6 M_\odot$, with $R_{1/2} = 2.44$ pc and number of star particles within 50 pc is $N_\star(< 50 \text{ pc}) = 120,515$ star particles. The half-mass radius is of the same order to the present-day value reported by Baumgardt (n.d.) but smaller by 1 pc.

5 Discussion

Using hydrochemical simulations following Bekki (2013) and a composite yield set based on Ven-

tura et al. (2012) and Ventura et al. (2013), the candidate GC reproduces a clear $[\text{Mg}/\text{Fe}]$ – $[\text{K}/\text{Fe}]$ anticorrelation that is visually consistent, within the observational uncertainties, with the NGC 1786 abundances reported by Alvarez Garay et al. (2025).

The sAGB polluter model (D’Ercole et al. 2010; Calura et al. 2019) remains a possible explanation for the observed anticorrelation pattern. Stars that have existed before the hydrochemical simulation started remain clustered near the initial GMC abundance (approximately $[\text{K}/\text{Fe}] \approx 0.0$ and $[\text{Mg}/\text{Fe}] \approx 0.5$ in our setup). This indicates that, within this simulation setup, the Mg–K spread is produced by enrichment of the gas that forms the new stars, rather than evolution of the pre-existing stellar population. This supports the plausibility of an sAGB-pollution plus dilution scenario as a working explanation for the observed trend.

In the simulated $[\text{Mg}/\text{Fe}]$ – $[\text{K}/\text{Fe}]$ plane, the distribution is relatively clustered near $[\text{Mg}/\text{Fe}] \approx 0.2$ and $[\text{K}/\text{Fe}] \approx 0.4$, while the extreme tail towards $[\text{Mg}/\text{Fe}] \approx -0.3$ and $[\text{K}/\text{Fe}] \approx 0.6$ becomes progressively sparser. One plausible interpretation is that the most extreme enrichment conditions are realised less frequently in the simulation: In the AGB/sAGB scenario, the most chemically extreme stars are expected when the star-forming gas is least diluted and most dominated by (super-)AGB ejecta; as pristine gas accretion and mixing become more important, the resulting abundances move back towards less extreme values (D’Ercole et al. 2008; D’Ercole et al. 2011; A. D’Ercole et al. 2016; Calura et al. 2019). The sparse extreme tail therefore plausibly reflects a short-lived or spatially limited phase of weak dilution, rather than a uniform enrichment history (D’Ercole et al. 2011; Calura et al. 2019).

We were unable to reproduce the NGC 1786 observational bounds using any single yield set alone (e.g. Ventura et al. (2012), Ventura et al. (2013), and Iliadis et al. (2016)). Instead, the best agreement required a composite choice: we adopted yields from Ventura et al. (2013) for $5\text{--}6 M_{\odot}$ and from Ventura et al. (2012) for $6\text{--}10 M_{\odot}$.

The candidate cluster also has broadly rea-

sonable structural properties (surface-density morphology, radial profile, total stellar mass, and half-mass radius). Quantitatively, we find $M(R < 50 \text{ pc}) = 2.83 \times 10^6 M_{\odot}$ and $R_{1/2} = 2.44 \text{ pc}$. These values are broadly comparable to observational estimates for NGC 1786 reported by Baumgardt (n.d.) and the mass adopted from Vanaraj et al. (2021), but with two clear offsets: our enclosed mass is higher by roughly one order of magnitude, and our half-mass radius is smaller by about 1 pc. Both differences are plausible if the simulated GC represents a younger evolutionary stage than the present-day cluster, since it would have had less time to lose mass (stellar evolution and tidal stripping) and to expand dynamically. With this caveat, the candidate GC remains a useful model for follow-up simulations.

A key uncertainty is the nucleosynthetic yield input. Different published yield sets (for example Ventura et al. (2013) and Iliadis et al. (2016)) predict different Mg depletion and K production at fixed mass and metallicity, and our result is therefore sensitive to how yields are selected and combined, and to the assumed dilution with pristine gas. A systematic exploration of yield choices, dilution fractions, and star-formation timing is needed to test whether the match is unique or one of many possible solutions. Our need for a composite yield choice underscores that the apparent agreement is not yet a unique solution.

6 Conclusions

We identify a candidate globular cluster analogue in a hydrochemical simulation that produces a clear Mg–K anticorrelation in newly formed stars, while pre-existing stars remain near the initial composition. This demonstrates that an sAGB enrichment plus dilution framework can plausibly reproduce the qualitative Mg–K trend reported for NGC 1786.

The comparison remains limited by several systematics. First, the agreement in the Mg–K plane is qualitative and not yet quantified with a statistical metric. Second, the result is not unique because the enrichment pattern depends on uncertain nucleosynthetic yields

and on the dilution history of star-forming gas; in our tests, the best agreement required a composite yield choice rather than a single published yield set. The candidate cluster has a compact half-mass radius and a radial density profile qualitatively similar in shape to NGC 1786, but the enclosed mass within 50 pc is higher than commonly adopted literature values by about an order of magnitude.

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